# Formatting Instructions For NeurIPS 2025

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## Abstract

## 1 Introduction

In the high stakes world of algorithmic decision making, the outcomes of machine learning models can have significant impacts on individuals and society in areas such as healthcare, criminal justice, lending, hiring. In this contexts, fairness, is not just a desirable property but a critical requirement TODO: [ADD CITATIONS]. In particular, group fairness, which requires predictative behaviour remains equitable across demographically defined groups, has emerged as a key focus of research in this area.

Notwithstanding the significant progress in developing group fairness algorithms, most of the existing work assumes a stationary data distribution which, in the real world, rarely holds. Instead, data distributions evolve over time: policy shifts, seasonal effects and changes in societyal norms can lead to distribution shifts (often referred to as concept drift) that can undermine predictative performance and fairness guarantees.

While online learning algorithms have been created to preserve predictative accuracy, there has been less work on how to preserve fairness under concept drift. This is a critical gap, because fairness violations may remain undetected longer than accuracy degradation, disrupting the trustworthiness of machine learning systems and leading to significant harms.

To address this challenge it is important not only to detect distributional changes as they occur but, at the same time, characterize their impact on fairness and react accordingly. In this paper, we study fairnes-aware learning under concept drift in online settings and investigate how a novel framework for detecting and responding to concept drift can be used to preserve fairness guarantees over time.

## 2 Experiments

Tables 1–4 report per-scenario results on Adult with mean $\pm$ standard deviation across seeds from `files/experiments`. Aranyani results use 4 available seeds; ARF and RFR use 5 seeds.

Table 1: No drift scenario results on Adult (mean $\pm$ std over seeds)

| Model | Accuracy $\uparrow$ | DP $\downarrow$ |
|---|---|---|
| Aranyani | $\mathbf{0.8538 \pm 0.0035}$ | $0.0860 \pm 0.0022$ |
| ARF | $0.8350 \pm 0.0000$ | $0.1040 \pm 0.0000$ |
| RFR | $0.8038 \pm 0.0074$ | $\mathbf{0.0592 \pm 0.0071}$ |

Table 2: Abrupt gender scenario results on Adult (mean $\pm$ std over seeds)

| Model | Accuracy $\uparrow$ | DP $\downarrow$ |
|---|---|---|
| Aranyani | $\mathbf{0.8607 \pm 0.0172}$ | $\mathbf{0.0414 \pm 0.0167}$ |
| ARF | $0.8100 \pm 0.0000$ | $0.1154 \pm 0.0000$ |
| RFR | $0.8025 \pm 0.0043$ | $0.0639 \pm 0.0017$ |

Table 3: Gradual gender scenario results on Adult (mean $\pm$ std over seeds)

| Model | Accuracy $\uparrow$ | DP $\downarrow$ |
|---|---|---|
| Aranyani | $\mathbf{0.8573 \pm 0.0022}$ | $\mathbf{0.0129 \pm 0.0063}$ |
| ARF | $0.8400 \pm 0.0000$ | $0.0890 \pm 0.0000$ |
| RFR | $0.8063 \pm 0.0054$ | $0.0668 \pm 0.0106$ |

Table 4: Occupation gender reversal scenario results on Adult (mean $\pm$ std over seeds)

| Model | Accuracy $\uparrow$ | DP $\downarrow$ |
|---|---|---|
| Aranyani | $\mathbf{0.8516 \pm 0.0045}$ | $0.0854 \pm 0.0056$ |
| ARF | $0.8200 \pm 0.0000$ | $0.1154 \pm 0.0000$ |
| RFR | $0.7888 \pm 0.0096$ | $\mathbf{0.0753 \pm 0.0111}$ |

Table 5: Folktables dataset results (mean $\pm$ std over seeds)

| Model | Accuracy $\uparrow$ | DP $\downarrow$ |
|---|---|---|
| Aranyani | | |
| ARF | $0.735 \pm 0.0000$ | $0.0750 \pm 0.0000$ |
| RFR | $0.747 \pm 0.0175$ | $0.0257 \pm 0.0168$ |

## References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

## A Technical Appendices and Supplementary Material

Technical appendices with additional results, figures, graphs and proofs may be submitted with the paper submission before the full submission deadline (see above), or as a separate PDF in the ZIP file below before the supplementary material deadline. There is no page limit for the technical appendices.

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